25901237 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECECT202, MG4SECECC200 - SOLAR TECHNOLOGY AND APPLICATIONS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. The resistance is measured in Ohm and the symbol of Ohm is a Greek alphabet …..... [U] / [CO1]

   a). μ b). π

   c). Ω d). β

2. Regular monitoring of hybrid systems helps in: [U] / [CO2]

   a). Increasing module size b). Early fault detection

   c). Increasing load d). Reducing sunlight

3. What is the first step before starting any maintenance work in a solar power plant? [K] / [CO3]

   a). Switch off and isolate the system b). Clean the panels

c). Replace components d). Check energy output

4. A solar PV module gives high power when the intensity of light falling is …….. [U] / [CO1]

a). Low b). Minimum

c). High d). None of these

5. Overloading in a circuit causes: [U] / [CO2]

a). Voltage increase b). Excess heating

c). Reduced current d). Improved efficiency

6. During the discharging process, chemical energy stored in batteries gets …..... [K] / [CO1]

a). E148 b). D148

c). G148 d). Converted into electrical energy

7. MPPT in an inverter stands for [K] / [CO2]

a). Maximum Power Point Tracking b). Minimum Power Performance Test

c). Module Power Processing Tool d). Maximum Panel Protection Technique

8. Which factor improves social impact in solar PV projects? [U] / [CO3]

a). High panel temperature b). Local employment generation

c). Increased pollution d). Grid overload

9. 1 kWh is equal to …….. [U] / [CO1]

a). 36 k J b). 3600 k J

c). 36000 k J d). 360 k J

10. A charge controller prevents: [U] / [CO2]

a). Overcharging of battery b). Overheating

c). Power export d). Shading

11. Earthing inspection is important during maintenance…..... [U] / [CO3]

a). Protect against electrical shock b). Improve appearance

c). Increase energy generation d). Reduce dust

12. What is the role of an inverter in an off-grid system? [U] / [CO2]

a). Convert AC to DC b). Regulate charging current

c). Convert DC to AC d). Store energy

13. The parameter that helps to compare PV systems of different sizes is ….. [U] / [CO3]

a). Total energy b). Specific yield

c). Voltage d). Current

14. Suppose, a solar cell produce 1 A current under 1000 W/m2 input solar radiation, then under 500W/m2 input solar radiation, the cell will produce ….. [U] / [CO1]

a). 0.25 A b). 0.5 A

c). 1.0 A d). 2.0 A

15. The electricity generated by solar PV panels in EVs is stored in ….. [U] / [CO3]

a). Fuel tank b). Supercapacitor only

c). Motor windings d). Traction battery

16. The lifespan/cycle life of a ….. is significantly higher than that of an SMF battery. [U] / [CO1]

a). Zinc chloride batteries b). Tubular battery

c). Li-ion battery d). Magnesium cell batteries

17. In battery sizing, which unit is commonly used? [K] / [CO2]

a). Watt b). Ampere

c). Ampere-hour (Ah) d). Ohm

18. What is formed when light is absorbed in quantum dots? [U] / [CO3]

a). Neutrons b). Photons

c). Excitons d). Phonons

19. Charge controllers protect the …… [U] / [CO1]

a). Rectifier b). AC-AC converters

c). Battery d). DC chopper

20. Solar panel efficiency depends on: [U] / [CO2]

a). Sunlight intensity b). Wire length

c). Switch type d). Color of roof

21. Which property of OPV cells makes them suitable for flexible applications? [U] / [CO3]

a). High temperature tolerance b). Lightweight and the ability to bend

c). High efficiency d). Large thickness

22. The primary function of batteries in a hybrid system is to: [U] / [CO2]

a). Increase irradiance b). Increase shading

c). Reduce panel size d). Store excess energy

23. Which condition can lead to battery undercharging in a PV system? [U] / [CO3]

a). Proper MPPT operation b). Faulty charge controller

c). High irradiance d). Correct wiring

24. The most important parameter of a PV module is its …..... [K] / [CO1]

a). Peak impedance b). Peak reverse impedance

c). Peak inverse output power d). Peak output power

25. A structurally weak roof may lead to: [U] / [CO2]

a). Higher efficiency b). Increased shading

c). Installation failure d). Improved cooling

26. Which type of motor is commonly used with DC solar water pumps? [U] / [CO3]

a). Stepper motor b). Synchronous motor

c). DC motor d). Induction motor

27. Which cable is used between PV modules and inverter? [K] / [CO2]

a). Armored AC cable b). Coaxial cable

c). Ethernet cable d). DC solar cable

28. In a solar PV system the main purpose of tilt angle optimization is to ….. [U] / [CO3]

a). Improve panel appearance b). Maximize energy generation

c). Reduce installation cost d). Prevent shading

29. The photovoltaic effect involves the absorption of photons, which generates ….. [K] / [CO1]

a). Electron - hole pairs b). Electron - neutron pairs

c). Electron - proton pair d). Electron - electron pair

30. A photovoltaic (PV) cell, commonly called a solar cell, is a ….. that converts sunlight directly into electricity. [U] / [CO1]

a). Acousto-optic devices b). Magneto-optic devices

c). Electro-mechanical device d). Non-mechanical device

31. Name the fundamental building block of solar photovoltaic technology ….. [U] / [CO1]

a). Electrochemical cell b). SMF cell

c). Tubular cells d). Solar cell

32. Inverter efficiency mainly affects: [U] / [CO2]

a). DC generation b). Panel temperature

c). Solar radiation d). AC output

33. Which emerging trend helps reduce land usage for solar plants? [U] / [CO3]

a). Ground-mounted solar PV systems b). Floating solar PV systems

c). Diesel hybrid solar PV systems d). Fixed tilt solar PV systems

34. In a voltaic cell (battery), two half-cells are connected by a …..... [U] / [CO1]

a). Salt bridge b). F157

c). D157 d). PV cell

35. Which type of data is collected by solar PV monitoring systems? [K] / [CO3]

a). Weather data only b). Electrical data only

c). Both electrical and environmental data d). Mechanical data

[1x35 = 35]

Part B

Short Essay Questions

Answer any **3** questions

Each question carries **5** marks

36. Describe thin-film solar PV material and their advantages and disadvantages. [K] / [CO1]

37. Describe the safety precautions that should be followed during solar panel installation. [U] / [CO2]

38. Explain the role of an inverter in a solar PV system. [U] / [CO1]

39. How does real-time monitoring benefit solar PV system performance? [U] / [CO3]

40. Write down the uses and advantages of a tubular battery. [K] / [CO1]

[5x3 = 15]